A BRIEF HISTORY OF

OCEAN ACOUSTICS

"OCEAN ACOUSTICS IS HIGHLY INTERTWINED WITH THE PHYSICAL, BIOLOGICAL, CHEMICAL, AND GEOLOGICAL ASPECTS OF THE OCEAN THAT DEFINE THE FIELD OF OCEANOGRAPHY"



1826

The first measurement of the speed of sound in water occurs in Lake Geneva by Jean-Daniel Colladon and Charles Sturm.

1490

Early evidence of ocean acoustics is found in Leonardo da Vinci's writings, where he describes how sound travels in water.

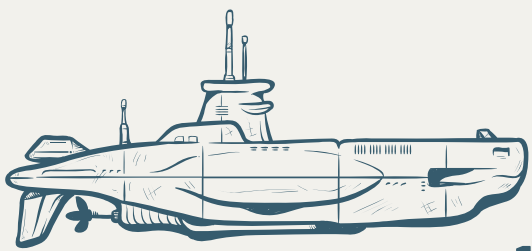


1914

Reginald Fessenden develops early sonar, detecting an iceberg that was 450 feet long and 130 feet high above the waterline from more than two miles away.

1918

World War I sees the damaging effects of German submarine attacks on shipping between the United States and Europe, leading to the widespread use of ocean acoustics tools for anti-submarine warfare.



1929

The Acoustical Society of America (ASA) is founded as part of the American Institute of Physics, enabling the acoustics community to gather and share research.



1945

The end of World War II sees oceanographic laboratories shifting from wartime efforts to basic research in oceanography, including ocean acoustics.



1970

The National Oceanic and Atmospheric Administration (NOAA) is established. Funding for oceanographic laboratories increases with the onset of the International Decade of Ocean Exploration.

1980

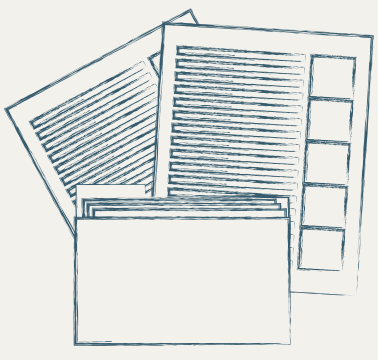
During the height of the Cold War, increased Naval support drives the growth of ocean acoustics programs in academia and industry, fostering innovation and drawing significant numbers of people into the field.

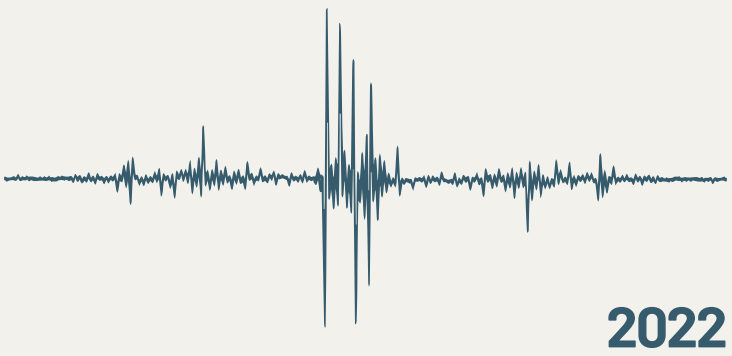
1997

The "Lackie Report," or Report of a Survey of U.S. Academic Programs in Ocean and Underwater Acoustics, is published, highlighting the need for more research funding in ocean acoustics and the designation of ocean acoustics as a National Naval Responsibility (NNR).

2000

Passive Acoustic Monitoring (PAM) emerges as a vital tool for marine conservation, allowing researchers to monitor marine mammal populations and assess the impact of anthropogenic noise on marine ecosystems.





2020

Ocean sound is added to the list of Essential Ocean Variables (EOVs) for the Global Ocean Observing System (GOOS), under both Physics/Climate and Biology/Ecosystems categories.

2022

The Office of Naval Research (ONR) sponsors the National Academies to conduct a study evaluating the state of ocean acoustics education and expertise, considering the growing number of federal and international ocean initiatives.

COMMITTEE ON OCEAN ACOUSTICS EDUCATION AND EXPERTISE, OCEAN STUDIES BOARD, DIVISION ON EARTH AND LIFE STUDIES, & NATIONAL ACADEMIES OF SCIENCES, ENGINEERING, AND MEDICINE. (2024). *OCEAN ACOUSTICS EDUCATION AND EXPERTISE* (P. 27337). NATIONAL ACADEMIES PRESS.